

1. Administrative & Study Identification

Full Study Title: Study to Investigate the Effect of Renal Impairment on the Pharmacokinetics, Safety, and Tolerability of AZD6234 **Short Title/Acronym:** AZD6234 Renal Impairment PK Study **Protocol Number:** D8750C00007 **NCT Number:** NCT06845813 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD6234 **Phase of Development:** Phase I **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, tolerability, and pharmacokinetics (PK) of a single subcutaneous dose of AZD6234 in participants with end-stage renal disease (ESRD), severe, moderate (optional), and mild (optional) renal impairment, and to compare them with matched participants having normal renal function. **Secondary Objectives:** To evaluate additional pharmacokinetic parameters, and the incidence of adverse events (AEs) and serious adverse events (SAEs) following the single-dose administration of AZD6234.

2.2 Background & Rationale To evaluate the impact of varying degrees of renal impairment on the pharmacokinetics, safety, and tolerability of the investigational drug AZD6234. Understanding the PK profile in renally impaired patients is critical for determining appropriate dosing recommendations and safety margins for its future clinical use.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Clinical Trial) **Control Type:** Matched healthy control group (matched by sex, age, and BMI) **Blinding:** Open-label **Randomization:** Non-randomised (Parallel-group assignment based on renal function)

3.2 Planned Enrollment & Duration Target \$N\$: 24 evaluable participants **Number of Sites:** Multiple US-based research locations (e.g., Orlando, FL; Lake Forest, CA) **Estimated Study Start:** 01-Jan-2025 **Estimated Primary Completion:** 30-Sep-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant must be 18 to 85 years of age, inclusive, at the time of signing the informed consent. 2.

Body weight ≥ 50 kg; BMI within the range of ≥ 18 to ≤ 40 kg/m², as measured at screening. 3. Stable renal function in the 3 months prior to dosing, with varying degrees of impairment assessed via BSA-adjusted CKD-EPI equation, or eGFR ≥ 90 mL/min for matched healthy controls.

4.2 Exclusion Criteria 1. History or presence of significant GI disease, previous upper GI surgery, or clinically significant cardiovascular disease. 2. Severe vitamin D deficiency < 12 ng/dL, or neuromuscular/neurogenic disease. 3. History or presence of any condition known to interfere with absorption, distribution, metabolism, or excretion of drugs.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose **Route of Administration:** Subcutaneous (SC) injection **Duration of Treatment:** Single administration under fasted conditions, with a follow-up period of up to 43 days (6 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Stable dose of cardio-renal relevant treatment for at least 2 weeks prior to screening. Stable management of controlled diabetes mellitus is also permitted. **Prohibited:** Oral contraceptives (due to potential effect of the IMP on drug absorption); phosphate binders, cholestyramine/colestipol, and ranitidine/nizatidine within 10 hours before and 10 hours after study intervention.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to -3 Informed Consent, Medical History, eGFR/BSA assessments, matching criteria review
Baseline	Day -1 to Day 1	Admission to Clinical Unit, Baseline labs, Single SC Dose of AZD6234, Inpatient PK profiling	
Interim	Days 8, 15, 22, 29, 36	Weekly outpatient visits: Safety Labs, ECGs, blood sample collection for PK, and AE recording	
End of Study	Day 43 (Week 6)	Final Follow-up Visit, Exit Interview, Final safety assessments	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Pharmacokinetics (PK) of AZD6234 and the number of participants experiencing Adverse Events (AEs) and Serious Adverse Events (SAEs) from Screening until the Follow-up visit (Day 43). **Measurement Method:** Clinical observation, vital signs, ECGs, and blood plasma concentration analysis.

7.2 Secondary & Exploratory Endpoints - Detailed evaluation of PK parameters including area under the curve (AUC) and maximum concentration ($C_{\max}$). - Tolerability profile across specific sub-groups (mild, moderate, severe impairment, and ESRD).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collected via resident observation during the inpatient period and weekly outpatient visits. **Serious Adverse Events (SAEs):** Evaluated strictly from Screening through to the End of Study follow-up (Day 43). **Data Monitoring Committee (DMC):** Internal safety review overseen by AstraZeneca clinical monitoring groups.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all AE/SAE evaluations and Pharmacokinetic (PK) Analysis Set for evaluating drug exposure across varying renal function groups. **Sample Size Justification:** $N=24$; standard for Phase I single-dose PK studies to adequately characterize systemic exposure across matched renal impairment and healthy control cohorts. **Handling of Missing Data:** Standard non-compartmental PK analysis rules apply (e.g., plasma concentrations below the limit of quantification treated as zero).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous inpatient monitoring followed by scheduled weekly outpatient medical reviews of safety data. **Audit Trail:** Standardized eCRF locking and safety data clarification protocols.

11. Study Identifiers & Governance

Primary ID: {{D8750C00007}} **Registry Number:** {{NCT06845813}}
Sponsor/Lead Organization: {{AstraZeneca}} **Study Title:** {{Study to Investigate the Effect of Renal Impairment on the Pharmacokinetics, Safety, and Tolerability of AZD6234}} **Official Regulatory Title:** {{Study to Investigate the Effect of Renal Impairment on the Pharmacokinetics, Safety, and Tolerability of AZD6234}}

12. Clinical Framework (Renal Impairment Specific)

Primary Condition: {{Nephropathy / Renal Impairment}} Diagnosis Threshold: {{eGFR based on BSA-adjusted CKD-EPI equation}} Medical Methodology: {{Pharmacokinetic & Safety Evaluation}} Optimization Target: {{AZD6234 PK and Safety Profiling}}

13. Study Procedures & Methodology

Management Pathway: {{Phase 1 PK/Safety profiling pathway}} Education Protocol (HCP): {{Protocol-specific dosing and safety monitoring training}} Education Protocol (Patient): {{Inpatient dosing requirements and outpatient visit onboarding}} Quality Audit Mechanism: {{Clinical site monitoring and strict AE/SAE tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{43 Days (approx. 6 Weeks) post-dose}} Onsite Visit Cadence: {{Days 8, 15, 22, 29, 36, and 43}} Remote Monitoring Frequency: {{N/A (Routine Outpatient assessments)}} Checklist Review Interval: {{Continuous during Day -3 to Day 5 inpatient stay}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				